Elias Weston-Farber

Research Engineer — Scientific AI & Clinical Trial Data Infrastructure

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GitHub: github.com/eliaswestonfarber **ORCID:** 0000-0002-8452-9432 **Publications:** PubMed Bibliography

Research engineer building AI systems for scientific evidence synthesis and clinical trial data infrastructure. Created ScienceAI, an open-source agentic harness that coordinates frontier language models for systematic literature analysis with provenance tracking. 9 peer-reviewed publications including NEJM and JAMA. BSPH AI Research Day Staff Award, 2026.

Publications & Awards

9 peer-reviewed publications (2 authored, 7 as METRC consortium collaborator — lead data engineer at the coordinating center), including **NEJM** and **JAMA • 4 manuscripts in press / under review** (METRC) **• 1 DOI-archived open-source software package** (10.5281/zenodo.21938772)

Full publication list: PubMed Bibliography

BSPH AI Research Day Staff Award — Johns Hopkins Bloomberg School of Public Health, March 2026. Awarded for podium presentation and poster session on ScienceAI alongside Jeff Leek's keynote. Selected from submitted abstracts across BSPH. Abstract co-authors: Renan Castillo, Katherine Frey.

Selected Projects

ScienceAI — Open-Source Agentic Harness for Scientific Literature Analysis *(2024–present)* - Designed and built a multi-agent orchestration system for systematic review and meta-analysis tasks: PI orchestrator delegates to Analyst Agents that perform structured data extraction with reflection/validation - Supports GPT, Claude, and Gemini as interchangeable backends with unified provider abstraction; field-level provenance tracking links every extracted data point to source quotes, page locations, and derivation explanations - Validated by reproducing Scolaro et al. (2014) meta-analysis: three frontier models independently recovered the primary finding (OR 2.15–2.37 vs. published 2.32) from 19 source papers - Published on PyPI (`pip install scienceai-llm`), GPL-3.0, 18 releases, full CI/CD, archived on Zenodo

Baltimore Comedy Place — Fully autonomous, AI-powered data pipeline that scrapes, cleans, and aggregates event listings from disparate sources to create Baltimore's first comprehensive comedy show guide

EliasTechLabs — Serverless personal site built on AWS Lambda, CloudFront, and S3, featuring an LLM-powered website assistant

RSmartsheet — R package interfacing with Smartsheet API, used by health authorities for project management

SearchIt — Systematic web search application used by NGOs for public data gathering and analysis

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Work Experience

Senior Cloud Engineer & Data Manager

METRC, Johns Hopkins Bloomberg School of Public Health *2023 – Present*

- Manage cloud infrastructure and data pipelines powering multi-site randomized controlled trials across 70+ Level I trauma centers nationwide
- Engineered an automated DSMB reporting system for nine studies, replacing a manual approach that required five full-time analysts with one that now requires only a single analyst
- Decreased cloud costs by 20% by migrating parts of the backend to Compute Engine while simultaneously improving system responsiveness and user engagement through HTMX and WebSockets, enabling real-time feedback
- Implemented a real-time Firebase database to ensure data synchronization across distributed environments

Senior Research Application Developer & Data Manager

Johns Hopkins Bloomberg School of Public Health *2020 – 2023*

- Led and managed a team of six analysts and developers, successfully setting up seven studies on an analytic platform built on GCP, automating weekly reporting
- Designed and implemented a system for real-time study data issue tracking and notifications, replacing a manual static file-based system and facilitating rapid response measures that supported a major publication
- Developed tailored visualization and reporting solutions for research teams, integrating technologies such as Selenium and Scrapy to extract conflicts of interest data, and Plotly to create interactive reports

Research Application Developer

Johns Hopkins Bloomberg School of Public Health *2019 – 2020*

- Developed a robust administrative data storage solution for managing records across 30+ METRC studies, leveraging custom Smartsheet & Google Sheets APIs and a cross-platform Electron application

- Wrote multiple R packages and designed a Python+R Docker environment to standardize the computational environment for METRC study analysis

Backend Developer & Data Scientist

Civicly Involved, Chicago *2018 – 2019*

- Led a team of interns in designing and implementing API-based data integration and management solutions, including geospatial data analysis using Google Cloud technologies

Education

University of Maryland Baltimore County, Baltimore — *2018 Bachelor of Science in Environmental Science & Geography Bachelor of Arts in Political Science*

Technical Skills

- **Languages:** Python, R, JavaScript, SQL
- **AI/ML:** Multi-agent LLM orchestration, provider API integration (OpenAI, Anthropic, Google), structured data extraction, prompt engineering, evaluation design
- **Frameworks:** Flask, Node.js, PyTorch, Pandas, Tidyverse, Shiny, Electron, HTMX
- **Infrastructure:** Docker, AWS, GCP, GitHub Actions, Firebase, WebSockets
- **Data & Analysis:** Tableau, ArcGIS, QGIS, Plotly, ggplot2